

Emissions and accumulation of metals in the atmosphere due to crackers and sparkles during Diwali festival in India

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Abstract

In India, Diwali is known as one of the most famous festivals. On the occasion of this festival, people burn crackers and sparkles to express their happiness. The burning of these fireworks leads to metal pollution in air. In this study, metal concentrations in ambient air were observed to be very high as compared to background values on previous days. For some metals the concentrations were observed to be higher than reported at industrial sites. The order of concentration of metals on the day of festival was observed to be in the order— $K > Al > Ba > Mg > Fe > Sr > Na > Ca > Cu > Mn > As > V > Ni > Bi$. Interestingly, the concentrations of Ba, K, Al and Sr went up to 1091, 25, 18 and 15 times higher than the previous day of Diwali. This study indicated that burning of crackers and sparkles on Diwali is a very strong source of air pollution which contributes significantly high amount of metals in air.

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1. Introduction

Accumulation of metals in the atmosphere is of great concern to all of us as they adversely affect human health and ecosystem if present in excess concentration. Despite the necessity of some of the metals in all living organisms, certain metals cause various toxic effects if accumulated in animal tissues (Yasutake and Hirayama, 1997). There are various types of sources emitting these metals into the atmosphere, e.g. fossil fuel combustion contributes Al, Fe, Ca, Mg, K, Na, As, Pb, Cd, Sc, Hg metals (Furimsky, 2000), metals like Pb and Zn are contributed by wood combustion (Mohn et al., 2002), vehicular traffic contributes Cd, Cr, Cu, Ni, Pb, Zn (Westerlund, 2001), electroplating contributes Cr (Flow-

er et al., 1994), metal alloy industries emit different metals like Cd, Cr, Al, Fe, Ni, Zn, Pb, Cu, etc. in the air (Harrison, 1986). One of the interesting source of the metals in the atmosphere in India is Diwali, one of the biggest festivals of India. During this festival crackers and sparkles are burnt emitting metal pollutants into the atmosphere in addition to other pollutants. A study carried out by Attri et al. (2001) showed the formation of O_3 without participation of NO_x due to burning of sparkles. This festival is celebrated over a span of 5 days. But a huge amount of crackers and sparkles are burnt mainly on the main day of the festival. During the year 2002, the main day of Diwali happened to be 4th November. In this study, we have measured the levels of metals emitted by the burning of crackers and sparkles during the festival period by collecting aerosol samples. This effort was made time over, to understand the air pollution problem related to metal concentrations in air on the occasion of Diwali.

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2. Experimental

2.1. Site and sample collection

Indian Institute of Chemical Technology (IICT) is located in north-east part of Hyderabad city (17.5°N, 78.5°E). Fig. 1 shows the location of sampling site. The locality of IICT from where aerosol samples were collected mainly represents residential area. The samples were collected at IICT terrace using aerosol sampler having 25 μm cut off diameter. This sampler has been designed and developed by us. Samples were collected on polytetrafluoroethylene (PTFE) filters with a flow rate of 25 LPM. The sampler was run for a sufficient time of about 8 h in order to pass around 10 m³ air through the filter. A total of five samples were collected, i.e. three samples before the festival (1–3 November, one

sample on the festival day (4 November) and one sample after the festival day (5 November).

2.2. Sample preparation and analysis

Half of the filter (aerosol sample) was extracted in 5 ml of 5% nitric acid (Fluka Trace Select Ultra grade) by keeping in ultrasonic bath for 1 h. The acid was diluted by using trace pure water (Fluka). The extracted solution was aspirated into ICP-MS for analysis. A reagent blank was also prepared and analysed. Samples were analysed quantitatively using multielement standards (Fluka, Switzerland). A three point calibration was used for quantification. The samples were analysed by Inductively Coupled Plasma- Mass Spectrometer (ICP-MS) (Varian Ultramass 700). The optimized

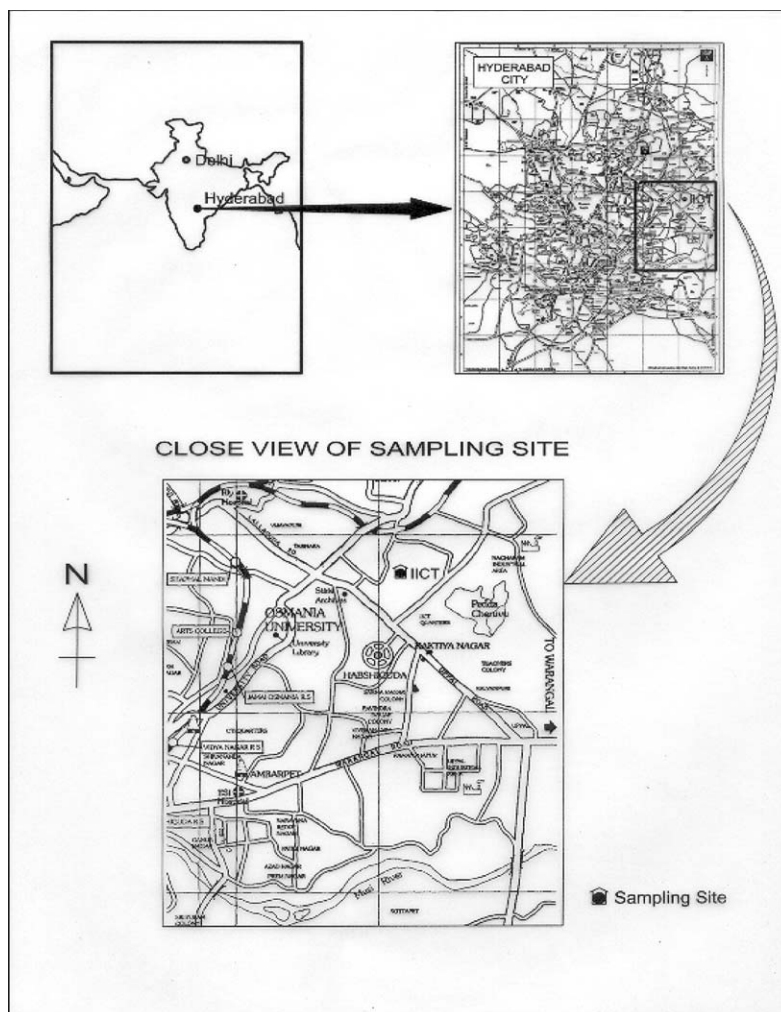


Fig. 1. Location of sampling site.

Table 1
ICP-MS instrumental conditions for analysis

Parameter	Set value
Plasma flow	15.5 lmin ⁻¹
Nebulizer flow	0.95 lmin ⁻¹
Sampling depth	6.5 mm
Auxilliary flow	1.25 lmin ⁻¹
Pump rate	25 rpm
Power	1.2 kW
Scan type	Segmented
Spacing	0.1 AMU
Dwell time	200 μ s
Scan per replicate	25
Replicate per scan	5

Table 2
Detection limits of measurements of metals by ICP-MS

Metal	Detection limits (ngl ⁻¹)
Al	13
As	10
Ba	0.6
Bi	0.5
Ca	10
Cu	3
Fe	3
K	5
Mg	1
Mn	7.9
Na	15
Ni	23
Sr	0.8
V	1.2

instrumental conditions are given in Table 1. Threshold values of detection by ICP-MS are given in Table 2.

3. Results and discussion

3.1. General variations

Variations of different metal levels during Diwali festival days have been shown in Figs. 2a–c. Fig. 2a shows the variation of Al, Ba, K, Mn, Fig. 2b shows the variation of Fe, Na, Sr, Ca, Cu and Fig. 2c shows the variation of Bi, Ni, As, V. In general, all the metals have very sharp peaks on 4th November, i.e. the day of the festival because tremendous amount of crackers and sparkles were burnt on this very day. Before and after the festival, the levels are much lower. Ba, K, Al, As, Sr, Cu, Mg, Ni, V are observed to be higher by a factor of 1091, 25, 18, 16, 15, 9, 6, 4, 1.5, respectively, while Bi

and Na were observed three times higher followed by Ca, Fe and Mn which were two times higher. Sharp peaks on 4th November clearly indicate the contribution of burning of fireworks and sharp decrease of these metals on the next day suggest their accumulation for short period followed by deposition on to the ground.

The highest concentration 58 $\mu\text{g m}^{-3}$ was observed for K followed by Al, Ba, Mg, Fe, Sr, Na, Ca, Cu, Mn, As, V, Ni, Bi. It is to be noted that 6.3, 4.7 and 0.7 $\mu\text{g m}^{-3}$ Al, K, and Mg, respectively, have been reported at an industrial site at Agra by Kulshrestha et al. (1994) which are much lower than reported on the day of Diwali (Fig. 1). Similarly, concentrations of Cu, Mn, Al, K and Mg are very high on the day of the festival as compared to the values reported at urban site in India (Tripathi et al., 2004). Sparkles and crackers mainly contain black powder that contains potassium nitrate, charcoal and sulphur (<http://www2.truman.edu/~blamp/chem121/project/06/06.html>). K salts such as chlorates, perchlorates and nitrates are used to supply oxygen necessary for combustion of the mixture. Metals like Al and Mg are used to make the crackers more bright and colourful. Elements such as Sr, Na, Cu and Ba are used to give red, yellow, blue and green colours, respectively. Fe helps to produce spark while Ca helps to deepen the colour produced by other elements. Similarly, rest of the metals are also used in fireworks to give more colouring effects and flash.

3.2. Correlations

Table 3 gives the correlation coefficients of various metals determined in aerosol samples. Interestingly, all metals show very strong correlations indicating that their source is the same, i.e. burning of crackers and sparkles during Diwali period. It is to be noted that K, Sr and Ba have perfect correlation with many metals (Table 3). It may be due to the abundant use of salts of these metals in fireworks.

4. Conclusion

This study shows that the burning of crackers and sparkles on the occasion of Diwali is a strong source of metals in ambient air. Some of the metals are emitted in very high quantity, as high as 1091 times for Ba, 25 times for K, 18 times for Al and 15 times for Sr as compared to one day before the festival. But the metal concentrations decrease sharply within the next 24 h indicating their accumulation for few hours. The concentration of K was observed to be highest (58 $\mu\text{g m}^{-3}$) on the day of the festival. The possible reason for highest K concentration is that potassium salts like chlorates, perchlorates, nitrates are the basic

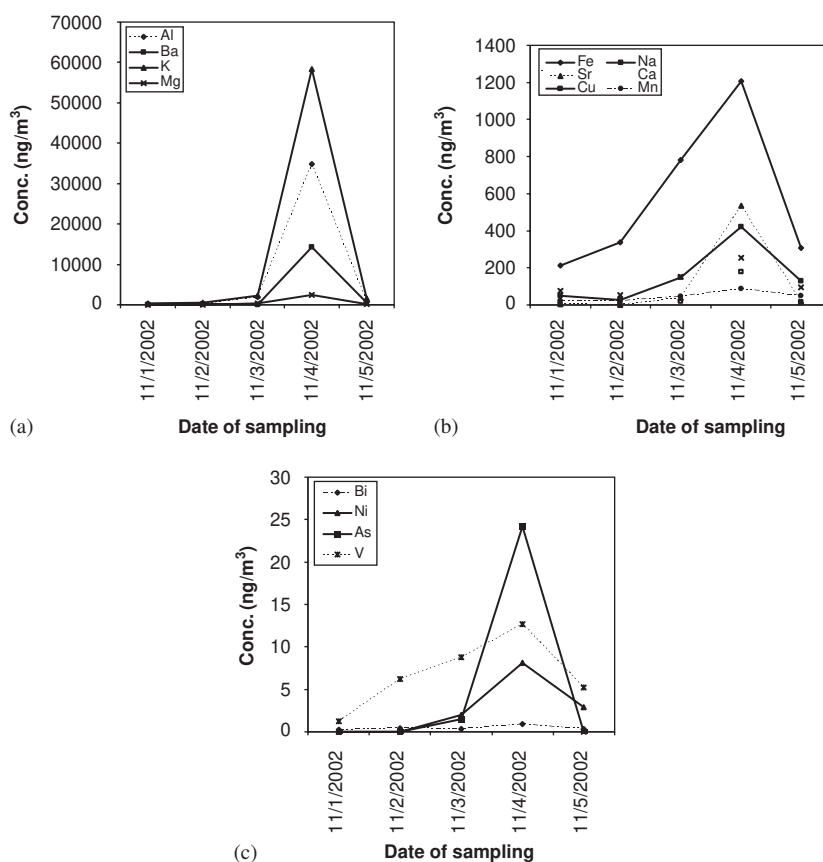


Fig. 2. (a) Variation of Al, Ba, K and Mg during the festival. (b) Variation of Fe, Sr, Cu, Na, Ca and Mn during the festival. (c) Variation of Bi, Ni, As and V during the festival.

Table 3
Correlation matrix of metals in aerosol samples collected at Diwali

	Al	As	Ba	Bi	Ca	Cu	Fe	K	Mg	Mn	Na	Ni	Sr	V
Al	1.00													
As	1.00	1.00												
Ba	1.00	0.99	1.00											
Bi	0.90	0.89	0.89	1.00										
Ca	0.91	0.93	0.89	0.75	1.00									
Cu	1.00	1.00	1.00	0.91	0.92	1.00								
Fe	0.89	0.91	0.87	0.82	0.96	0.90	1.00							
K	1.00	1.00	1.00	0.90	0.90	1.00	0.88	1.00						
Mg	0.99	0.99	0.98	0.85	0.96	0.99	0.92	0.99	1.00					
Mn	0.91	0.92	0.91	0.87	0.93	0.93	0.88	0.91	0.93	1.00				
Na	0.88	0.90	0.88	0.61	0.93	0.88	0.82	0.88	0.93	0.84	1.00			
Ni	0.92	0.92	0.92	0.87	0.89	0.94	0.83	0.92	0.93	0.98	0.85	1.00		
Sr	1.00	1.00	1.00	0.91	0.91	1.00	0.90	1.00	0.99	0.92	0.88	0.93	1.00	
V	0.78	0.78	0.76	0.92	0.75	0.80	0.86	0.78	0.76	0.82	0.50	0.74	0.80	1

materials used in the firework industries. Significantly higher concentrations of other metals may be due to their use in crackers and sparkles for giving colouring effects.

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